

①. Data understanding : col data types, null values, outliers.

1 Data cleaning

- ↓
- null values handle.
 - outlier handling.
 - duplicate removal.

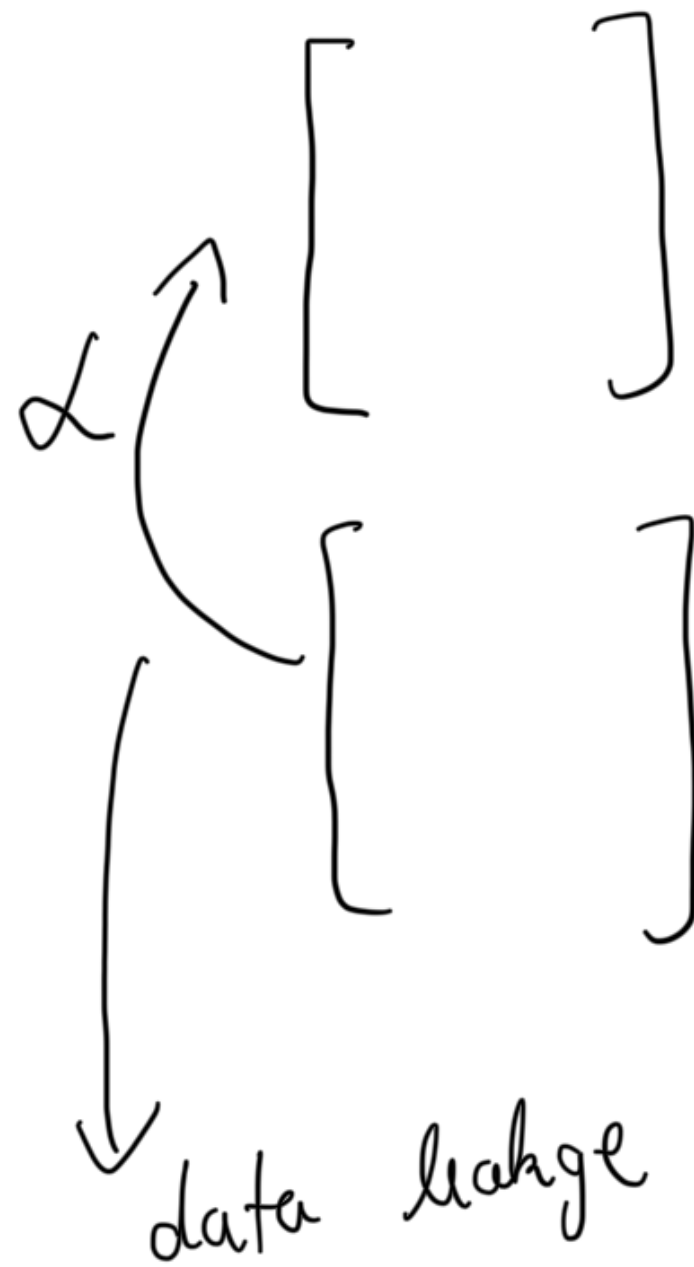
1.1 : Data processing / feature engineering.

ordered categories → Label encoding.

unordered → one-hot encoding.



recall, precision

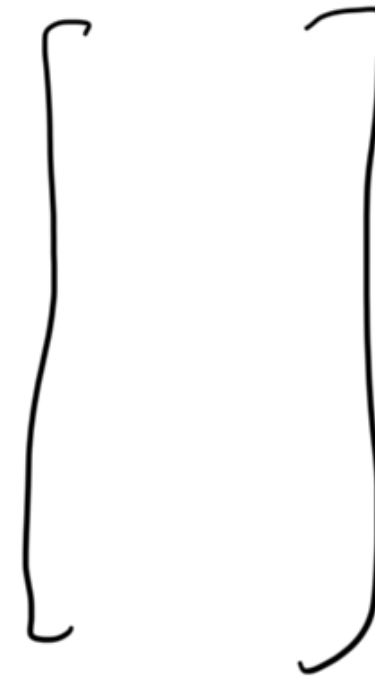


train

test

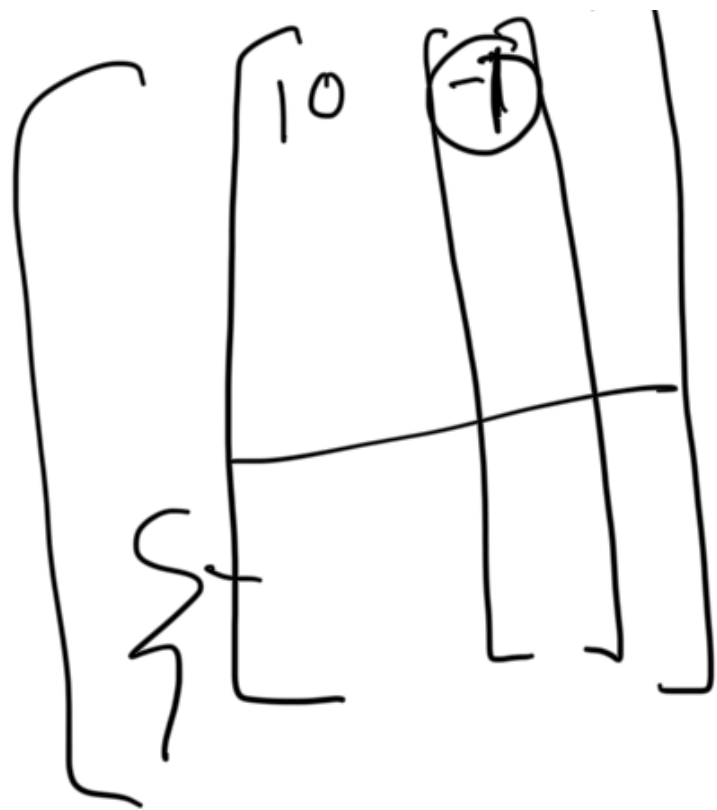
independent.

scale, fit, transfer (X)



train test split

scaling

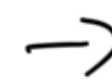


$$\left[\frac{x_i - \mu}{s.d} \right]$$

$$[] - \mu$$



Scaling



model

train ✓
val ✓



performance

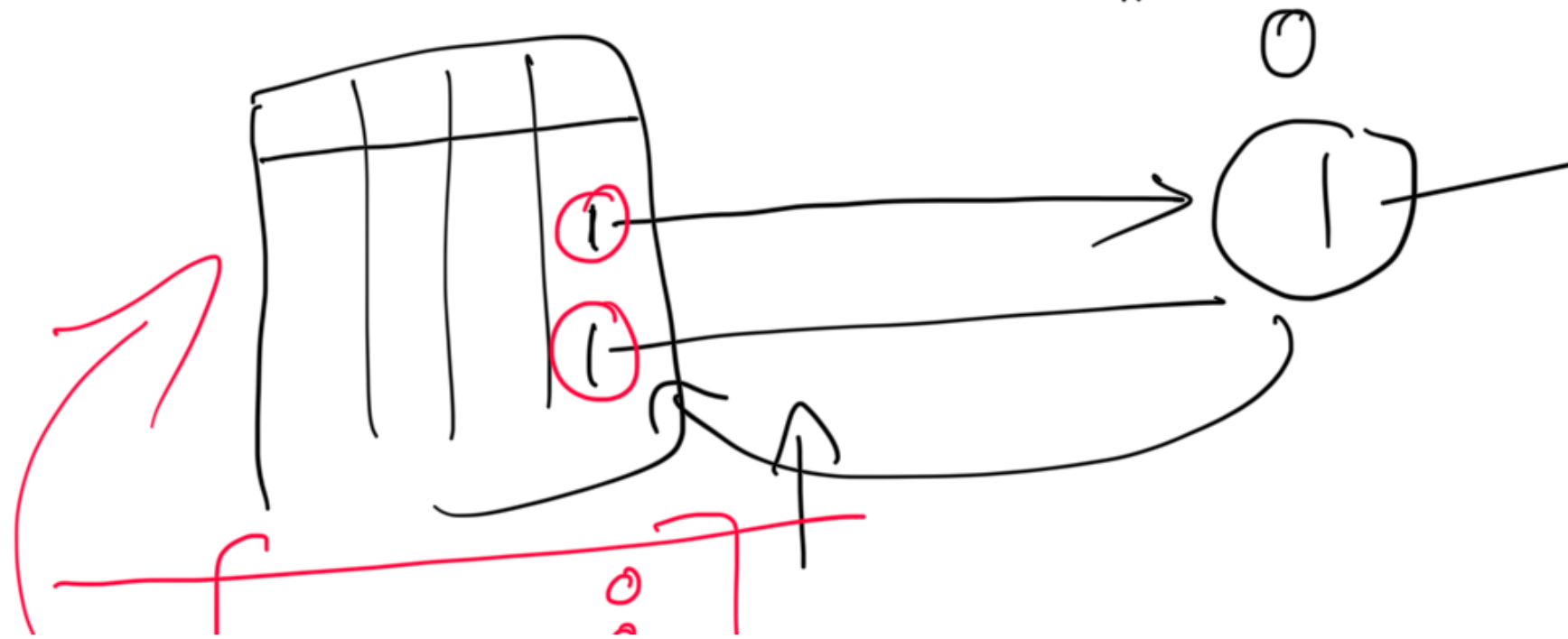
() []

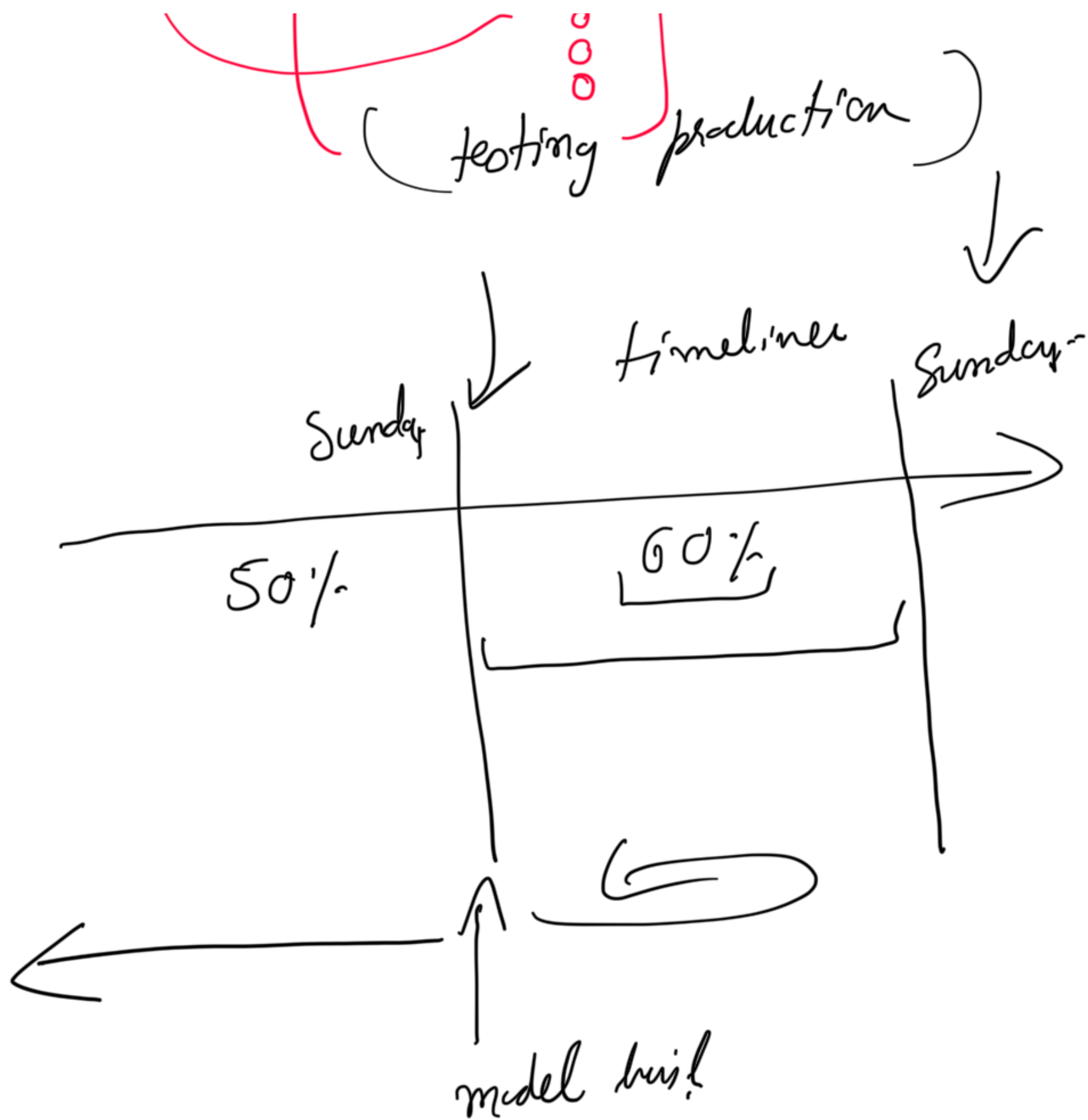
[] → train (historical)

[] → testing → future.

user_blocked

fraud





... Hand → 6 may ~

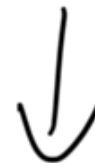
data

60%

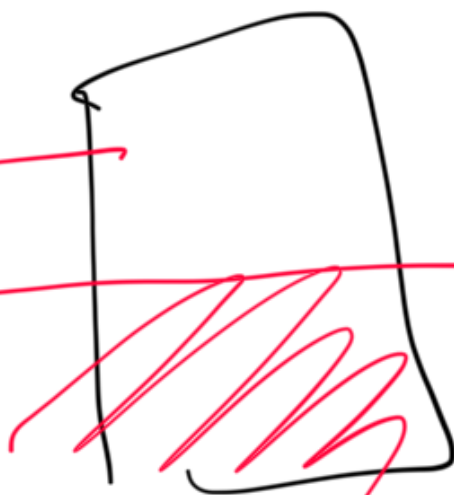
S-May.

SMOTE

class - imbalance



undersampling



→ CompuX

→ statquest :

↓
SMOTE
Synthetic minority
oversampling tech.